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## Method Article

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# Comparative Study of Data-Level Imbalance Handling Techniques with Ensemble Models for Credit Card Fraud Detection

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## Abstract

Credit card fraud detection remains a challenging problem due to extreme class imbalance, where fraudulent transactions constitute only a small fraction of total activity. This study presents a comparative analysis of six widely used data-level imbalance handling techniques; Random Undersampling, TomekLinks, Random Oversampling, SMOTE, SMOTETomek, and SMOTEENN paired with three ensemble classifiers: Random Forest, XGBoost, and LightGBM. Experiments were conducted on a large, publicly available synthetic transaction dataset using both 5-fold cross-validation and an 80/20 train-test split. Performance was assessed using AUC-ROC, F1-score, precision, and recall. XGBoost with TomekLinks achieved the highest F1-score of 0.8763 and an AUC of 0.9986, outperforming all other combinations. A paired t-test comparing F1-scores across sampling methods showed that XGBoost significantly outperforms LightGBM ( $p = 0.013$ ), with a mean F1-score difference of 0.1502. These results highlight XGBoost as a strong candidate for fraud detection under class imbalance, particularly when paired with TomekLinks resampling. The findings underscore the importance of selecting appropriate resampling strategies and suggest that model choice can be informed by both statistical evidence and practical deployment considerations.

**Keywords:** Credit Card Fraud Detection, Data-Level Imbalance Handling, Machine Learning Models, Class Imbalance Solutions, Fraud Prevention Techniques, Random Oversampling, SMOTE, Financial Security, Digital Payment Systems, Algorithmic Accuracy.

## 1 Introduction

Technology has significantly impacted financial transactions, promoting the use of cashless and contactless payments through platforms like fintech and e-commerce. This introduces risks, as the broad adoption and diverse user base heighten the likelihood of fraud. Credit and debit card fraud, whether stemming from theft, deceit, or the financial struggles of the cardholder, poses serious threats to financial security and stability [1].

The widespread use of cards, particularly among younger consumers now included in banking, increases the need for effective fraud prevention and detection strategies. Quick detection and management of fraud are essential to prevent financial and reputational damage to both consumers and financial institutions. Prevention strategies aim to halt suspicious transactions before they occur, while detection strategies focus on identifying fraud after it has happened [2].

Modern technology, particularly machine learning, plays an important role in enhancing fraud detection capabilities. These tools manage large volumes of transactions with greater

accuracy than human monitoring can achieve. Machine learning uses various methods; supervised, unsupervised, semi-supervised, and reinforcement learning to identify patterns and rules in data, enabling predictions on unknown data points.

In the context of credit card fraud detection, supervised learning proves especially useful as it requires classifications of transactions as either fraudulent or legitimate, while unsupervised learning treats fraud detection as anomaly detection, identifying fraudulent transactions as outliers.

This paper explores how different data-level techniques address the challenge of class imbalance in fraud detection datasets, where legitimate transactions vastly outnumber fraudulent ones. Techniques like Random Undersampling (RUS), Random Oversampling (ROS), and combinations such as Smote (Synthetic Minority Over-sampling Technique) + Tomek are employed to mitigate this imbalance.

The structure of this paper includes a comprehensive literature review of existing methods for managing data imbalance in fraud detection in Section 2, followed by a detailed overview of the dataset in Section 3. Data Processing and Exploratory Data Analysis are illustrated in Sections 4 and 5. Methodologies used in the study in 6. respectively. Results and analysis of this comparative analysis of various imbalance handling techniques are displayed in Section 7, which aim to highlight the most effective methods to enhance the accuracy and reliability of fraud detection systems. Data limitations are presented in 8. Conclusions and future work are discussed in Section 9.

The conclusion synthesizes these insights, providing recommendations for future research to further improve credit card fraud detection frameworks.

## 2 Literature Review

Credit card fraud detection presents a unique challenge due to the high imbalance between legitimate and fraudulent transactions. This class imbalance leads to skewed datasets that hinder the effectiveness of machine learning (ML) models, often biasing predictions towards the majority class [3, 4]. Consequently, a considerable body of research has focused on addressing this issue through various data-level techniques.

Several studies have explored strategies to mitigate class imbalance, underscoring the necessity of robust data handling techniques to enhance fraud detection performance. These works have documented a progression from basic resampling techniques to hybrid approaches. Early efforts included simple under-sampling and over-sampling methods, such as the Synthetic Minority Over-sampling Technique (SMOTE), which generates synthetic minority class instances to improve class balance [5].

Under-sampling techniques like Random Under-Sampling (RUS) and Tomek Links Under-sampling reduce the size of the majority class to match the minority class, though these methods risk discarding valuable data [4, 6]. On the other hand, over-sampling methods enhance the minority class, with SMOTE interpolating between minority instances to create synthetic data [3].

Hybrid techniques combine the strengths of both under-sampling and over-sampling. For instance, SMOTE-ENN and SMOTE-Tomek integrate noise removal with synthetic sampling to balance the datasets more effectively [3].

Comparative analyses such as those conducted in recent literature demonstrate that hybrid methods generally outperform isolated resampling techniques in improving detection accuracy. These studies evaluate performance using metrics such as Accuracy, Precision, Recall, F1-Score, and the Area Under the Curve (AUC), as reported by [7].

As the field has matured, ensemble learning methods have emerged as powerful tools for fraud detection. Techniques such as Random Forests, Gradient Boosting (including XGBoost), and LightGBM combine multiple classifiers to enhance detection accuracy and reduce false positives [5, 8, 9]. These ensemble models are valued for their robustness, capacity to handle imbalanced data, and strong generalization capabilities.

By focusing on the comparative performance of established data-level techniques and ensemble machine learning classifiers, this paper aims to contribute to the development of more accurate and reliable fraud detection systems. The insights gained from these approaches are particularly relevant in financial contexts where reducing false negatives is crucial for minimizing fraud-related losses.

### 3 Dataset Description

The dataset described in this research encompasses simulated credit card transactions spanning two years, from January 1, 2019, to December 31, 2020. The dataset is obtained from Kaggle, and its link is <https://www.kaggle.com/datasets/kelvinkelue/credit-card-fraud-prediction/data>. The dataset includes a total of 555,719 transaction records featuring 22 attributes, alongside a target column where a '1' indicates a fraudulent transaction and '0' signifies a legitimate one. This dataset encapsulates activities from 1000 unique customers engaging in transactions with 800 different merchants. A description of the dataset features is displayed in Table 1.

Several studies have utilized this Kaggle dataset. For instance, recent research by [10] applied various machine learning algorithms such as Logistic Regression, SVM, Random Forest, XGBoost, AdaBoost, and LightGBM achieving near perfect accuracy using XGBoost after preprocessing and cross-validation. Similarly, a study on decision tree-based fraud detection confirmed the relevance of this dataset for evaluating classification models in financial fraud contexts [11].

**Table 1:** Description of Dataset Features

| Feature Name          | Data Type            | Description                                                               |
|-----------------------|----------------------|---------------------------------------------------------------------------|
| id                    | Numeric - Discrete   | Unique identifier for each transaction.                                   |
| Trans_date_trans_time | DateTime             | Timestamp of the transaction (date and time).                             |
| Cc_num                | Numeric - Discrete   | Unique customer identification number.                                    |
| Merchant              | String               | Name of the merchant involved in the transaction.                         |
| Category              | Categorical          | Type of transaction (e.g., personal, childcare).                          |
| Amt                   | Numeric - Continuous | Transaction amount.                                                       |
| First                 | String               | Cardholder's first name.                                                  |
| Last                  | String               | Cardholder's last name.                                                   |
| Gender                | Binary               | Cardholder's gender.                                                      |
| Street                | String               | Cardholder's street address.                                              |
| City                  | Categorical          | Cardholder's city of residence.                                           |
| State                 | Categorical          | Cardholder's state of residence.                                          |
| Zip                   | Categorical          | Cardholder's zip code.                                                    |
| Lat                   | Numeric - Continuous | Latitude of cardholder's location.                                        |
| Long                  | Numeric - Continuous | Longitude of cardholder's location.                                       |
| City_pop              | Numeric - Discrete   | Population of the cardholder's city.                                      |
| Job                   | Categorical          | Cardholder's job title.                                                   |
| Dob                   | Date                 | Cardholder's date of birth.                                               |
| Trans_num             | String               | Unique transaction identifier.                                            |
| Unix_time             | Numeric - Discrete   | Transaction timestamp in Unix format.                                     |
| Merch_lat             | Numeric - Continuous | Latitude of the merchant.                                                 |
| Merch_long            | Numeric - Continuous | Longitude of the merchant.                                                |
| Is_fraud              | Binary               | Indicator of whether the transaction is fraudulent (1) or legitimate (0). |

### 4 Data Preprocessing

To ensure the dataset was suitable for analysis and model training, several preprocessing steps were conducted. These steps were essential to convert raw transactional data into a structured and model-ready format. The preprocessing procedure focused on data cleaning, feature transformation, categorical grouping, and encoding.

Initially, columns deemed irrelevant for classification were removed. These included `id`, `cc_num`, `merchant`, `first`, `last`, `street`, `city`, `zip`, `trans_num`, and `unix_time`, as they held identifiers or redundant information that would not enhance model performance.

The `trans_date_trans_time` column was converted to a datetime format, enabling the extraction of temporal features such as `month`, `day`, and `hour`. The `year` feature was excluded

because all transactions occurred in 2020. Additionally, the `day` feature was used to derive the `weekday` of each transaction, after which it was dropped.

To incorporate age as a feature, the `dob` (date of birth) column was used to calculate the cardholder’s age at the time of the transaction. The original `dob` column was subsequently discarded. Geolocation features including `lat`, `long`, `merch_lat`, and `merch_long` were used to compute the great-circle distance (in kilometers) between the cardholder and the merchant. These original latitude and longitude columns were then removed.

Categorical consolidation was applied to simplify and enhance feature interpretability. The `category` column was grouped into 14 broader categories (e.g., entertainment, food and dining, gas transport). Similarly, the `job` column, originally containing 16 distinct job titles, was recategorized into 9 broader industry groups such as business, healthcare, and education. The `state` column was classified into 4 U.S. geographic regions: North, Midwest, South, and West.

For numeric feature normalization, continuous variables including `month`, `hour`, `age`, `distance`, `amt`, and `city_pop` were standardized to ensure comparability and improve model convergence. Categorical variables specifically `weekday`, `category`, `gender`, `state`, and `job` were one-hot encoded to facilitate model compatibility. Notably, the `gender` column was binarized, with 0 representing male and 1 representing female.

Upon completion of the preprocessing steps, the final dataset consisted of 555,719 records and 42 columns, including the target variable `is_fraud`. These transformations were critical in preparing the data for efficient and accurate machine learning, ensuring that all features were consistently scaled, structured, and encoded.

## 5 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted in two phases: before and after preprocessing, to uncover patterns associated with fraudulent behavior and to assess the structure and quality of the dataset. The initial phase focused on the raw data, identifying irregularities, outliers, and potential indicators of fraud, while the second phase was conducted after applying data cleaning and feature engineering techniques, allowing for deeper analysis of relationships and trends in a refined dataset.

The distribution of key numerical features, particularly `transaction amount` and `city population`, revealed substantial positive skewness as seen in Figures 1a and 1b.

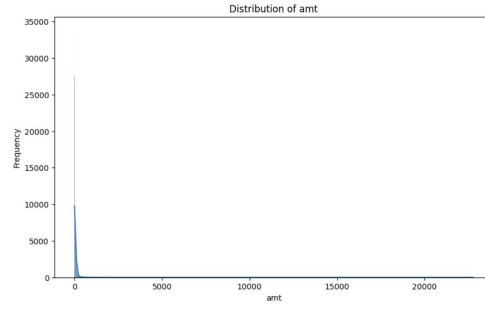
Most transactions involved small amounts, with only a few high-value transactions, and a majority of city populations were relatively low, with a few large urban centers. These characteristics justified the normalization of numerical features to minimize the influence of extreme values and improve model accuracy.

Correlation analysis among the limited set of numeric features showed minimal linear relationships among predictors. However, a slight positive correlation between `transaction amount` and the `is_fraud` target variable was observed, suggesting that larger transaction amounts may be more susceptible to fraud. Despite this weak correlation, the insight reinforces the importance of including transaction amount in predictive modeling. Correlations among numerical features are shown in the heatmap in Figure 2.

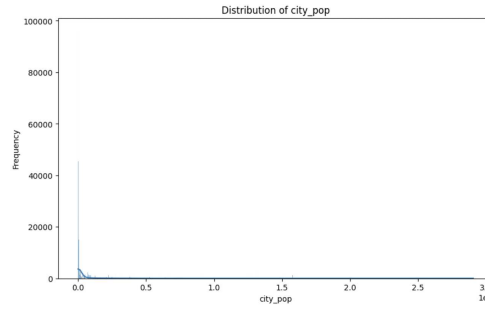
Analysis of nominal features provided insight into behavioral and demographic trends. Transaction types most commonly involved gas and transportation, groceries, and both physical and online retail purchases. A substantial number of cardholders held jobs in creative and technical fields such as video editing, design, architecture, and engineering. Geographically, Texas, New York, and Pennsylvania were the most represented states in terms of transaction volume. These trends offer contextual understanding of the dataset and potential high-risk segments.

Detailed inspection of fraudulent transactions by demographic and categorical attributes revealed that females participated in slightly more fraud cases (1,164) than males (981). Fraud occurred most frequently in the online and grocery shopping categories, with 506 and 485 instances, respectively. Occupations with notable fraud activity included science writers and licensed conveyancers. States such as New York (175 cases), Pennsylvania (114), and Texas (113) recorded the highest numbers of fraudulent transactions. These results can be viewed in Figure 3.

Following pre-processing, further EDA revealed temporal patterns of fraudulent activity. August had the highest incidence of fraud (415 cases), followed by October (384) and September (340), while June reported the lowest (133). Analysis by hour showed a spike in fraudulent transactions between 22:00 and 23:00, with 550 and 538 cases respectively, and significantly

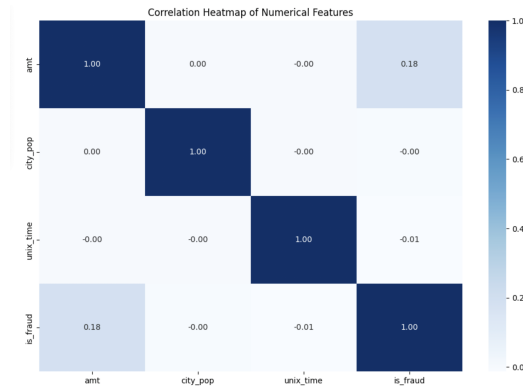


(a) Transaction Amount Distribution

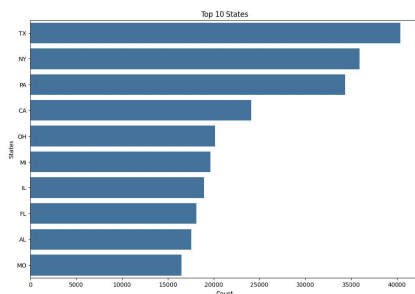


(b) City Population Distribution

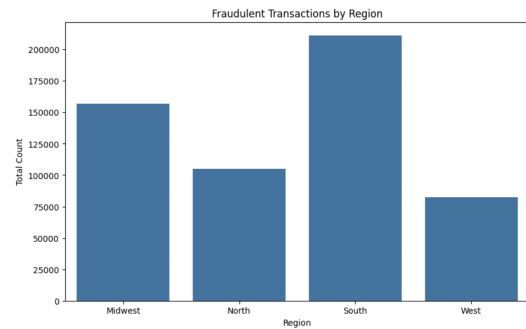
**Figure 1:** Skewed Distributions of Transaction Amount and City Population



**Figure 2:** Correlation Heatmap among Numerical Features



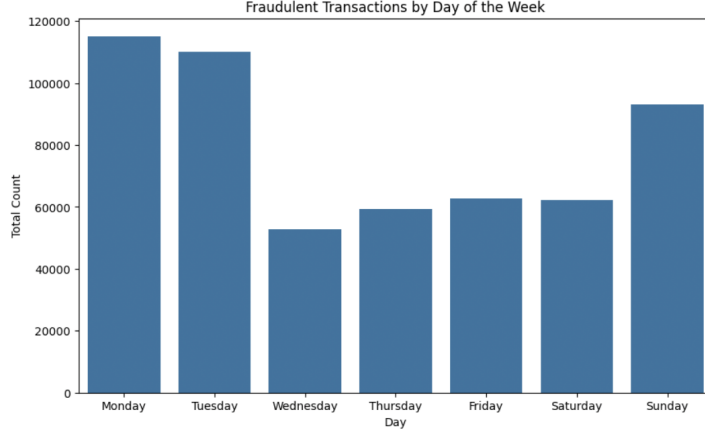
(a) Top 10 States in Transactions Fraudulent



(b) Fraudulent Transactions by Region

**Figure 3:** Transactions Fraudulent by States and Regions

fewer during early morning hours (e.g., 03:00 with only 194 instances). Spatial and temporal analysis revealed that the Southern region of the U.S. experienced the highest fraud rates, followed by the Midwest, North, and West. Additionally, Mondays, Tuesdays, and Sundays saw the most fraudulent activity, suggesting heightened risk during the beginning and end of the week. This can be seen in Figure 4.



**Figure 4:** Top Days in Fraudulent Transactions

These findings provide essential insight into how fraudulent behavior varies across demographic, geographic, and temporal dimensions. The results of the EDA directly informed preprocessing choices and model development, supporting a more targeted and data-driven approach to credit card fraud detection.

## 6 Methodology

This study investigates the effectiveness of combining various machine learning models with data imbalance handling techniques to improve credit card fraud detection. Fraud detection presents inherent challenges, notably extreme class imbalance and high-dimensional input data. To address these, we employed an integrated approach utilizing ensemble classifiers and diverse resampling strategies aimed at enhancing model performance and generalization.

The core classification task is binary, distinguishing between fraudulent and legitimate transactions. Logistic regression was first selected to serve as a baseline model due to its simplicity and widespread use. In addition, we utilized ensemble models known for their superior performance in similar tasks. These included Random Forest, which uses multiple decision trees to mitigate overfitting and increase robustness; Extreme Gradient Boosting (XGBoost), which iteratively improves performance through boosting techniques particularly effective in imbalanced contexts; and LightGBM, a scalable gradient boosting framework optimized using techniques like Gradient-based One-Side Sampling.

Given the small proportion of fraudulent cases in real-world datasets, it was critical to employ data-level resampling techniques to balance the training data. This study examined six strategies. For undersampling, we used Random Undersampling (RUS), which reduces the majority class by randomly removing instances, and Tomek Links, which identifies borderline examples to refine class boundaries without full balancing. Oversampling approaches included Random Oversampling (ROS), which duplicates minority class instances, and the Synthetic Minority Oversampling Technique (SMOTE), which generates new minority examples through interpolation. We also employed hybrid methods: SMOTE combined with Tomek Links (Smote-Tomek) and with Edited Nearest Neighbors (SmoteENN), enhancing both boundary refinement and noise removal.

These resampling techniques were applied to the training portion of the data, while the testing set remained untouched to allow for fair performance evaluation. Evaluation metrics included accuracy, precision, recall, F1-score, and the area under the ROC curve (AUC). However, due to the dataset’s imbalance, the focus was placed on precision, recall, and F1-score, as accuracy may misleadingly favor the majority class.

For data partitioning, the original dataset was divided into an 80%-20% training-test split while preserving the original class ratio. The test set contained 111,144 transactions and remained highly imbalanced, with only 0.386% labeled as fraudulent. The training dataset initially comprised 444,575 records, which were then transformed into six separate versions using the resampling techniques mentioned earlier. These versions varied in size and class balance. Four approaches; RUS, ROS, SMOTE, and SmoteTomek yielded balanced datasets. SmoteENN provided an almost balanced dataset, while Tomek Links preserved the original imbalance but reduced the dataset size for clearer class separation. The characteristics of these datasets are summarized in Table 2.

Recognizing the limitations of single train test split evaluations, we adopted a more robust approach using 5-fold cross-validation to assess model generalization. This method ensured that each resampled dataset underwent rigorous and unbiased performance testing across multiple folds, providing a comprehensive comparison of model performance under different imbalance handling techniques. Each fold maintained the same class ratios and sizes described in Table 2, although total sample sizes exceeded those of the initial training split.

The experimental setup was implemented in Python using the Scikit-learn and imbalanced-learn libraries. Care was taken to apply resampling only to training data within each fold to prevent data leakage. Ultimately, the objective of this methodology was to identify the most effective pairing of machine learning model and resampling technique, offering practical insights for the design of high-performing fraud detection systems.

**Table 2: Description of Dataset Features**

| Dataset              | Size (Transactions) | Imbalance Ratio (Legitimate : Fraud) |
|----------------------|---------------------|--------------------------------------|
| Original Dataset     | 555,719             | 99.614% : 0.386%                     |
| Random Undersampling | 3,432               | 50% : 50%                            |
| Tomek Links          | 444,338             | 99.614% : 0.386%                     |
| Random Oversampling  | 885,718             | 50% : 50%                            |
| SMOTE                | 885,718             | 50% : 50%                            |
| SmoteTomek           | 885,718             | 50% : 50%                            |
| SmoteENN             | 882,740             | 49.831% : 50.169%                    |
| Test Set             | 111,144             | 99.614% : 0.386%                     |

All classifiers were used with default hyperparameters provided by their respective libraries (scikit-learn, XGBoost, LightGBM), without grid search or tuning. This ensured a consistent baseline for evaluating the impact of resampling methods across models.

## 7 Results and Analysis

This section presents the evaluation of ensemble classifiers; Random Forest (RF), XGBoost (XGB), and LightGBM (LGBM), combined with six data-level imbalance handling methods: Random Undersampling, TomekLinks, Random Oversampling, SMOTE, SMOTETomek, and SMOTEENN. The models were trained and tested on a synthetic credit card fraud dataset using an 80/20 train-test split. Performance was assessed using AUC-ROC, accuracy, recall, precision, and F1-score.

Table 3 reports the results for each classifier-resampling combination. The AUC-ROC scores for nearly all models exceeded 0.99, indicating strong discriminative capability, especially for XGBoost and Random Forest under TomekLinks, SMOTE, and SMOTEEN. LightGBM, while achieving high recall across most resampling methods, tended to underperform in precision, particularly with Random Oversampling and SMOTE-based methods.

XGBoost with TomekLinks achieved the highest F1-score (0.87626), along with a high recall (0.80886) and precision (0.95592), highlighting its superior balance in detecting fraudulent transactions while minimizing false positives. Random Forest with SMOTE and SMOTEEN also performed well in terms of F1-score (0.81102 and 0.79697, respectively). LightGBM achieved its best F1-score with Random Undersampling (0.20207), though this is still notably lower than the other ensembles using hybrid or oversampling strategies.

To evaluate whether the difference in F1-score performance between XGBoost and LightGBM across the various resampling strategies was statistically significant, a paired t-test was conducted on the six F1-score values for each model. XGBoost achieved a mean F1-score of 0.7001 with a standard deviation of 0.2534, while LightGBM obtained a lower mean of 0.5499 and a standard deviation of 0.1867. The paired t-test returned a  $t$ -statistic of 3.75 with a  $p$ -value

**Table 3:** Performance Metrics of Ensemble Models with Various Resampling Techniques

| Model | Sampling Method      | AUC-ROC | Accuracy | Recall  | Precision | F1-Score       |
|-------|----------------------|---------|----------|---------|-----------|----------------|
| RF    | Random Undersampling | 0.99664 | 0.96958  | 0.96270 | 0.10932   | 0.19634        |
| XGB   | Random Undersampling | 0.99592 | 0.96781  | 0.98135 | 0.10549   | 0.19050        |
| LGBM  | Random Undersampling | 0.99673 | 0.97023  | 0.97669 | 0.11269   | 0.20207        |
| RF    | TomekLinks           | 0.99673 | 0.99852  | 0.62937 | 0.97826   | 0.76596        |
| XGB   | TomekLinks           | 0.99859 | 0.99912  | 0.80886 | 0.95592   | <b>0.87626</b> |
| LGBM  | TomekLinks           | 0.89471 | 0.99718  | 0.69697 | 0.62033   | 0.65642        |
| RF    | Random Oversampling  | 0.89471 | 0.99874  | 0.69697 | 0.96764   | 0.81030        |
| XGB   | Random Oversampling  | 0.99845 | 0.99760  | 0.92308 | 0.62857   | 0.74788        |
| LGBM  | Random Oversampling  | 0.99824 | 0.99162  | 0.95804 | 0.31042   | 0.46891        |
| RF    | SMOTE                | 0.99592 | 0.99870  | 0.72028 | 0.92793   | 0.81102        |
| XGB   | SMOTE                | 0.99734 | 0.99841  | 0.86247 | 0.75820   | 0.80698        |
| LGBM  | SMOTE                | 0.99694 | 0.99665  | 0.87646 | 0.54101   | 0.66904        |
| RF    | SMOTETomek           | 0.99694 | 0.99865  | 0.71329 | 0.91892   | 0.80315        |
| XGB   | SMOTETomek           | 0.99734 | 0.99841  | 0.86247 | 0.75820   | 0.80698        |
| LGBM  | SMOTETomek           | 0.99694 | 0.99665  | 0.87646 | 0.54101   | 0.66904        |
| RF    | SMOTEENN             | 0.99694 | 0.99855  | 0.73660 | 0.86813   | 0.79697        |
| XGB   | SMOTEENN             | 0.99733 | 0.99802  | 0.86713 | 0.69533   | 0.77178        |
| LGBM  | SMOTEENN             | 0.99664 | 0.99606  | 0.88345 | 0.49413   | 0.63378        |

of 0.013, indicating a statistically significant difference at the 5% level. The 95% confidence interval for the mean difference was [0.0473, 0.2531], confirming that XGBoost significantly outperforms LightGBM in terms of average F1-score across the tested resampling methods.

These results emphasize that model performance is highly sensitive to the choice of resampling technique. While undersampling methods achieved acceptable recall, their precision and F1-scores were consistently low due to class imbalance. In contrast, hybrid techniques like TomekLinks and SMOTEENN enabled better balance between sensitivity and specificity. Overall, XGBoost with TomekLinks stands out as the most effective configuration for this dataset, achieving the best F1-score while maintaining high AUC, precision, and recall.

## 8 Data Limitations

While the study presents a comprehensive comparative analysis of data-level imbalance handling techniques, several limitations inherent to the dataset and experimental design must be acknowledged. First, the dataset is characterized by an extreme class imbalance, with fraudulent transactions constituting only 0.386% of the total records. This severity imposes challenges on model training and evaluation, as standard classifiers may become biased toward the majority class despite resampling interventions. Moreover, the dataset used in this study is synthetically generated rather than derived from real-world banking systems. Although it mirrors the structure and complexity of genuine credit card transaction data, the synthetic nature of the dataset may limit the generalizability of the findings to operational environments where user behavior, transaction variability, and fraud patterns may differ significantly.

Another limitation concerns the lack of temporal modeling. Although the data spans a two-year period, the models employed in this study do not explicitly account for temporal dependencies or evolving transaction sequences, which could be pivotal in capturing dynamic fraud behaviors. Additionally, the preprocessing strategy involved the exclusion or generalization of high-cardinality categorical features such as merchant name, customer job title, and geographic identifiers. While this was necessary for computational efficiency and to reduce model overfitting, it may have led to the omission of features potentially rich in discriminatory power for fraud detection.

Finally, given that the study focuses exclusively on data-level imbalance handling, other effective approaches; such as algorithm-level adjustments, cost-sensitive learning, and hybrid frameworks, were not explored. As a result, the insights derived from this research, though methodologically robust, should be interpreted with caution and seen as a foundation for more expansive and realistic evaluations involving live transactional data and more complex modeling strategies.

## 9 Conclusion and Future Work

In this study, we conducted a comparative analysis of six data-level imbalance handling techniques combined with three ensemble classifiers; Random Forest, XGBoost, and LightGBM,

for credit card fraud detection. Experimental results demonstrated that pairing ensemble models with hybrid sampling strategies such as TomekLinks and SMOTEENN leads to substantial gains in performance, particularly in recall and F1-score.

XGBoost with TomekLinks achieved the highest F1-score (0.8763) and AUC (0.9986) on the test set, outperforming other combinations. A paired t-test showed that this improvement over LightGBM with equivalent resampling was statistically significant ( $p = 0.013$ ). Nevertheless, all tested models performed well in terms of AUC, indicating strong discriminatory power across the board.

These findings suggest that both the choice of resampling strategy and classifier architecture significantly influence performance in highly imbalanced classification problems. While XGBoost emerged as the most effective under the tested conditions, final model selection should also consider computational efficiency, interpretability, and scalability in deployment environments.

Future work could extend this analysis to additional real-world datasets, explore cost-sensitive and semi-supervised learning approaches, and evaluate the impact of concept drift in evolving transaction streams.

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## Competing Interests

The authors declare no competing interests.

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## Author Contribution

Noha Youssef supervised the research project and led the transformation of the original thesis into a scientific manuscript. Malak Gaballa, Masa Tantawy, and Moustafa El Mahdy carried out the data preprocessing, experimental design, and analysis, and were responsible for drafting the original thesis. All authors contributed equally to the conceptual development, interpretation of results, and finalization of the study.

## Data Availability Statement

The dataset used in this study is publicly available on Kaggle: <https://www.kaggle.com/datasets/kelvinkelue/credit-card-fraud-prediction/data>.

## Research Involving Human and/or Animals

Not Applicable.

## **Informed Consent**

Not Applicable.